

Midterm Exam 1 · Fall 2021

MATH 411 · October 1, 2021

Complete worked solutions

Retypeset past exam for practice. The original was closed book and worth 25 points.

1. Let $a, b, c, d \in \mathbb{R}$ satisfy $ad - bc = 1$, and set

$$f(z) = \frac{az + b}{cz + d}, \quad \mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}.$$

- (a) **(2 points)** Show that f maps $\mathbb{H}$ into $\mathbb{H}$.

Solution. Since $cz + d \neq 0$ for $z \in \mathbb{H}$ (its zero, if any, is real), we have

$$f(z) = \frac{az + b}{cz + d} = \frac{(az + b)(c\bar{z} + d)}{|cz + d|^2} = \frac{ac|z|^2 + bc\bar{z} + adz + bd}{|cz + d|^2}.$$

So if $z = x + iy$ and $\bar{z} = x - iy$, then

$$\Im f(z) = \frac{y(-bc + ad)}{|cz + d|^2} = \frac{y}{|cz + d|^2} > 0,$$

where we have used the fact that $ad - bc = 1$ and also that $y > 0$.

- (b) **(2 points)** Is $f : \mathbb{H} \rightarrow \mathbb{H}$ holomorphic?

Solution. As both $az + b$ and $cz + d$ are holomorphic (they are polynomials, hence entire), the ratio is also holomorphic whenever $cz + d \neq 0$. If $c \neq 0$, the only zero of $cz + d$ is $z = -d/c$, which is real. If $c = 0$, then $d \neq 0$ since $ad - bc = 1$. Thus $cz + d$ does not vanish anywhere on $\mathbb{H}$.

- (c) **(5 points)** A line or circle can be given by an equation

$$Ax + By + C(x^2 + y^2) = D,$$

where $A, B, C, D \in \mathbb{R}$ and $z = x + iy$. For the choice $(a, b, c, d) = (0, -1, 1, 0)$, show that $f(z) = -1/z$ sends the portion of such a line or circle in $\mathbb{H}$ to the portion of a line or circle in $\mathbb{H}$.

Solution. Note that if $z = x + iy$, then

$$f(z) = -1/z = \frac{-x}{x^2 + y^2} + i \frac{y}{x^2 + y^2}.$$

Let $u(x, y) = \frac{-x}{x^2 + y^2}$ and $v(x, y) = \frac{y}{x^2 + y^2}$. Note that

$$u^2 + v^2 = \frac{1}{x^2 + y^2}.$$

If x, y satisfy $Ax + By + C(x^2 + y^2) = D$, then also

$$A \frac{x}{x^2 + y^2} + B \frac{y}{x^2 + y^2} + C = D \frac{1}{x^2 + y^2}.$$

Rewriting this in terms of u and v we obtain:

$$-Au + Bv + C = D(u^2 + v^2).$$

Rearranging this, we find that u, v satisfy

$$(-A)u + Bv + (-D)(u^2 + v^2) = -C,$$

which is the equation of a circle/line in the $u - v$ plane. Since $f(f(z)) = z$ for $z \in \mathbb{H}$, this equation describes precisely the image in $\mathbb{H}$.

2. Consider the real function

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

- (a) **(3 points)** Show that f is infinitely differentiable at 0 and compute every $f^{(k)}(0)$. You may use that a rational function of x multiplied by e^{-1/x^2} tends to 0 as $x \rightarrow 0$.

Solution. By definition

$$f'(0) = \lim_{h \rightarrow 0} \frac{e^{-1/h^2} - f(0)}{h} = \lim_{h \rightarrow 0} \frac{e^{-1/h^2}}{h} = 0$$

where we have used the hint as well as the fact that $f(0) = 0$. Now, we proceed inductively assuming that $f^{(k)}(0) = 0$ and show that $f^{(k+1)}(0) = 0$ as well. Indeed, by definition

$$f^{(k+1)}(0) = \lim_{h \rightarrow 0} \frac{f^{(k)}(h) - f^{(k)}(0)}{h}.$$

Now for $x \neq 0$, the function e^{-1/x^2} is certainly differentiable, since e^x is differentiable, $-1/x^2$ is differentiable, and we can use the chain rule. In fact, one can easily verify by induction on k that there exist polynomials $p_k(x), q_k(x)$ such that, for $h \neq 0$, $q_k(h) \neq 0$ and we have

$$f^{(k)}(h) = \frac{p_k(h)}{q_k(h)} e^{-1/h^2}. \quad (1)$$

The case $k = 0$ has $p_0 = q_0 = 1$. Indeed, suppose the claim is true for $j = k - 1$. Then by the product rule,

$$f^{(k)}(h) = \frac{d}{dh} \left(\frac{p_{k-1}(h)}{q_{k-1}(h)} \right) e^{-1/h^2} + \frac{2}{h^3} \frac{p_{k-1}(h)}{q_{k-1}(h)} e^{-1/h^2},$$

again for $h \neq 0$. But the derivative of a rational function is again a rational function, so the inductive claim has been proven. Now use the hint and (1) to conclude that $f^{(k+1)}(0) = 0$ as well.

- (b) **(3 points)** Prove or disprove: an analytic function on a complex neighborhood of 0 agrees with f on the real axis near 0.

Solution. Suppose that $g(z)$ is an analytic function around 0 agreeing with $f(x)$ for points on the real axis with $|x|$ sufficiently small. By what we showed in class, if $\sum_{n \geq 0} a_n T^n$ is the formal series such that $g(z) = \sum_{n \geq 0} a_n z^n$ for z near 0, then $a_k = \frac{g^{(k)}(0)}{k!}$. As g agrees with f on points along the real axis, we may as well compute the $\mathbb{C}$ -derivatives of $g(z)$ by approaching 0 through real values. Hence, we are reduced to the computation from the previous part, and so $a_k = 0$ for all k . In other words, $g(z)$ must be 0 near 0, a contradiction since $f(x)$ is non-zero near 0.

3. (10 points) Suppose

$$f(T) = 1 + a_1T + a_2T^2 + \cdots$$

has positive radius of convergence and every a_k is a positive real number. The formal inverse $g(T) = \sum_{n \geq 0} b_n T^n \in \mathbb{C}[[T]]$ satisfies $f(T)g(T) = 1$. Prove that g also has positive radius of convergence.

Hint. First find a recurrence for b_n , then choose $0 < \delta < 1$ such that $\sum_{k \geq 1} a_k \delta^k < 1$.

Solution. From the relation $g(T)f(T) = 1$, if we collect the coefficients of T^n we get the following relations

$$T^0 : 1 \cdot b_0 = 1 \tag{2}$$

$$T^1 : b_1 + a_1 b_0 = 0 \tag{3}$$

$$T^2 : b_2 + a_1 b_1 + a_2 b_0 = 0 \tag{4}$$

$$\vdots :$$

$$T^n : b_n + a_1 b_{n-1} + \cdots + a_{n-1} b_1 + a_n b_0 = 0 \tag{5}$$

Solving for b_n in the last relation, we obtain that

$$b_n = - \sum_{k=1}^n a_k b_{n-k}. \tag{6}$$

Now write

$$f(T) = \sum_{n \geq 0} a_n T^n = 1 + h(T), \text{ so that } h(T) = a_1 T + a_2 T^2 + \cdots.$$

If $f(T) = 1$, then clearly $g(T) = 1$ has a positive radius of convergence and there is nothing to prove, so assume $f(T)$ is non-constant and hence $h(T) \neq 0$. Note that $f(T)$ and $h(T)$ have the same radius of convergence, denote this radius by r .

On $D(0, r)$, $h(z)$ is continuous, and $h(0) = 0$. So by continuity we can choose $0 < \delta < \min\{1, r\}$ such that on $\overline{D(0, \delta)}$ we have $|h(z)| < 1$. We shall contend that

$$|b_n| \leq \frac{1}{\delta^n},$$

which, by Hadamard, shows that $g(T)$ has radius of convergence at least δ .

We can prove the claim by induction, where $b_0 = 1$ so the $n = 0$ case is verified. Now assume the inductive claim for all b_j with $j < n$. Then we compute using (6) that

$$|b_n| \leq \sum_{k=1}^n |a_k| \frac{1}{\delta^{n-k}} = \frac{1}{\delta^n} \sum_{k=1}^n a_k \delta^k \leq \frac{1}{\delta^n} \sum_{k \geq 1} a_k \delta^k \leq \frac{1}{\delta^n},$$

where in the last line we have used the fact that a_j are real, positive and also that $\sum_{k \geq 1} a_k \delta^k = h(\delta)$.